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MC180400735: SADAQAT IQBAL AHMED

Time Left 80 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 30 of 30 (Start time: 10:10:49 AM, 02 July 2020)

Total Marks: 1

A function V is said to be harmonic if and only if _____.

Select the correct option

Reload Math Equations

☐	$V_x + V_y = 0$
☐	$V_{xy} + V_{yx} = 0$
☒	$V_{xx} + V_{yy} = 0$
☐	$V_x^2 + V_y^2 = 0$

Sir usama

correct

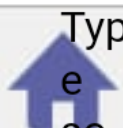
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MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 28 of 30 (Start time: 10:09:46 AM, 02 July 2020)

Total Marks: 1

If $f(z)=z^2$, then $f(x+iy)=$ _____.

Select the correct option

Reload Math Equations

☐	$x^2 - y^2 + 2xyi$	correct
☐	$x^2 + y^2 - 2xyi$	
☐	$x^2 + y^2i - 2xyi$	
☐	$x^2 - y^2i - 2xyi$	

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MC180400735: SADAQAT IQBAL AHMED

Time Left 78 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 23 of 30 (Start time: 10:08:01 AM, 02 July 2020)

Total Marks: 1

Given a complex - valued function $s(t)$, a rotation by an angle $\frac{\pi}{3}$ is given by _____ .

Select the correct option

Reload Math Equations



$$e^{\frac{3\pi}{3}} s(t)$$



$$e^{\frac{2\pi}{3}} s(t)$$



correct

$$e^{\frac{\pi}{3}} s(t)$$



none of the above

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MC180400735: SADAQAT IQBAL AHMED

Time Left 79 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 20 of 30 (Start time: 10:07:02 AM, 02 July 2020)

Total Marks: 1

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 then the $\lim_{z \rightarrow z_0} f(z)$ _____ .

Select the correct option

Reload Math Equations

- ☐ exists
- ☐ does not exist

correct

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MC180400735: SADAQAT IQBAL AHMED

Time Left 79 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 17 of 30 (Start time: 10:05:51 AM, 02 July 2020)

Total Marks: 1

The Cauchy - Riemann equations on a pair of real - valued functions of two real variables $u(x, y)$ and $v(x, y)$ are $U_x = V_y$ and _____.

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--------------|
| ☐ | $U_y = -V_y$ |
| ☐ | $U_x = -V_x$ |
| ☐ | $U_y = -V_x$ |
| ☐ | $U_x = -V_y$ |

correct

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1

4G



69%



10:04 a.m.

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MC180400735: SADAQAT IQBAL AHMED

Time Left

85
sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 15 of 30 (Start time: 10:04:47 AM, 02 July 2020)

Total Marks: 1

Let $z = 1 + i$, then $\arg(z) =$ _____.

Select the correct option

Reload Math Equations

 $2\pi/4$  $3\pi/4$  $4\pi/4$  $\pi/4$

correct

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Tabs



1

4G



69%



10:04 a.m.

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Time Left

84

sec(s)



MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 14 of 30 (Start time: 10:04:05 AM, 02 July 2020)

Total Marks: 1

$$\lim_{(x,y) \rightarrow (1,2)} 3xy^2 - y = \underline{\hspace{2cm}}$$

Select the correct option

Reload Math Equations



8



9



10



11

correct

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Time Left 79 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 11 of 30 (Start time: 10:01:19 AM, 02 July 2020)

Total Marks: 1

Product of complex numbers $(3 + 2i)$ and $(1+7i)$ is _____.

Select the correct option

- ☐ 3+9i
- ☐ 4+9i
- ☒ -11+23i
- ☐ 3+14i

correct

Click to Save Answer & Move to Next Question



70%



9:58 a.m.

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Time Left

83
sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 3 of 30 (Start time: 09:58:04 AM, 02 July 2020)

Total Marks: 1

Let $z = 7i$, then $\arg(z) =$ _____.

Select the correct option

Reload Math Equations

 $\pi/4$  $\pi/2$

correct

 $\pi/3$  $\pi/6$

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Tabs



70%



9:57 a.m.

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MC180400735: SADAQAT IQBAL AHMED

Time Left

83
sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 2 of 30 (Start time: 09:57:21 AM, 02 July 2020)

Total Marks: 1

By definition, $\lim_{z \rightarrow z_0} f(z) = L$ means that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that _____.

Select the correct option

Reload Math Equations



$$|f(z) - L| < \varepsilon \text{ whenever } 0 < |z - z_0| < \delta$$

correct



$$|f(z) - z_0| < \varepsilon \text{ whenever } 0 < |z - L| < \delta$$



$$|f(z) - L| < \delta \text{ whenever } 0 < |z - z_0| < \varepsilon$$



$$|f(z) - z_0| < \delta \text{ whenever } 0 < |z - L| < \varepsilon$$

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Suppose that $\lim_{z \rightarrow z_0} f(z) = A$ and $\lim_{z \rightarrow z_0} g(z) = B$, then choose the correct option .

Select the correct option

Reload Math Equations

☐	$\lim_{z \rightarrow z_0} [f(z) + g(z)] = A + B$	correct
☐	$\lim_{z \rightarrow z_0} [f(z) + g(z)] = A - B$	
☐	$\lim_{z \rightarrow z_0} [f(z) + g(z)] = A \cdot B$	
☐	$\lim_{z \rightarrow z_0} [f(z) + g(z)] = \frac{A}{B}$	

Click to Save Answer & Move to Next Question

(Start time: 06:37:30 PM, 01 July 2020)

The function $u(x, y) = \frac{xy}{x^2 + y^2}$ does not have a limit when _____

option

$(x, y) \rightarrow (1, 1)$

$(x, y) \rightarrow (0, 1)$

$(x, y) \rightarrow (1, 0)$

$(x, y) \rightarrow (0, 0)$

correct

Question # 23 of 30 (Start time: 06:48:57 PM, 01 July 2020)

Total Marks: 1

If two complex - valued functions f and g satisfy conditions $g(f(z)) = z$ and $f(g(w)) = w$, then these functions are _____ of each other.

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|-------------------|
| ☐ | conjugates |
| ☐ | inverses |
| ☐ | reciprocals |
| ☐ | none of the above |

correct



MC180201604: AZKA SALOMI

Time Left

90

sec(s)

MTH632:Grand Quiz

Quiz Start Time: 05:04 PM, 01 July 2020

Question # 17 of 30 (Start time: 05:11:02 PM, 01 July 2020)

Total Marks: 1

A complex-valued linear transformation is always _____ on the entire complex plane.

Select the correct option



only one-to-one



only onto



one-to-one and onto



none of the above

correct

Click to Save Answer & Move to Next Question

Question # 5 of 30 (Start time: 11:53:51 AM, 01 July 2020)

Total Marks: 1

In the complex valued function $z = x - iy$, the value of $\Re\{U_x\} = \underline{\hspace{1cm}}$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|----|
| ☐ | -1 |
| ☐ | 0 |
| ☐ | 1 |
| ☐ | 2 |

correct

$$k=0 = \sqrt{3} - i$$

$$-\sqrt{3} - i$$

$$k=2 =$$

$$2i$$

$$k=1 =$$

A complex-valued function of the form

where

$$L(z) = Az + B,$$

Select the correct option

☐	linear
☐	quadratic
☐	cubic
☐	none of the above

correct

A connected open set is called a _____.

Select the correct option

☒	domain
☐	range
☐	closed set
☐	unbounded set

correct





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MC180402225: ASMA BILAL

MTH632:Grand Quiz

Quiz St

Question # 18 of 30 (Start time: 09:49:03 AM, 01 July 2020)

The complex number

$$(4i)^4$$

have _____ number of roots.

Select the correct option



1



2



3



4

correct

Click to Save Answer



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MCT60402223: ASMA BILAL

MTH632:Grand Quiz

Question # 12 of 30 (Start time: 09:44:20 AM, 01 July 2020)

Let $z = 3$, then $\arg(z) =$ _____.

Select the correct option



0 degree

correct



60 degree



90 degree



270 degree



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MC180402225: ASMA BILAL

Time Left

87

sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:34 AM, 01 July 2020

Question # 16 of 30 (Start time: 09:46:47 AM, 01 July 2020)

Total Marks: 1

The Jordan Curve Theorem guarantees that a simple closed curve must enclose a region.

Select the correct option



Euler curve



Jordan curve



Simple curve



Rotation curve

Click to Save Answer & Move to Next Question



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MC180402225: ASMA BILAL

Time Left

88

sec(s)

MTH632:Grand Quiz

Quiz Start Time: 09:34 AM, 01 July 2020

Question # 15 of 30 (Start time: 09:46:09 AM, 01 July 2020)

Total Marks: 1

A complex-valued linear transformation could be a composition of

_____.

Select the correct option



translation



rotation



magnification



all of the above

correct

Click to Save Answer & Move to Next Question



MTH632:Grand Quiz

Question # 10 of 30 (Start time: 09:41:15 AM, 01 July 2020)

If $-8i = 8 \exp[i(-\pi/2 + 2k\pi)]$, then the root of

$$(-8i)^{1/3}$$

for $k=1$ is _____.

Select the correct option

- | | |
|----------------------------------|-----|
| ☐ | 3i |
| ☐ | -3i |
| ☒ | 2i |
| ☐ | -2i |

correct

Question # 28 of 30 (Start time: 08:49:02 AM, 01 July 2020)

Total Marks: 1

If $f(z) = z^3$, $g(z) = z + 2$, then $g(f(5)) =$

Select the correct option

[Reload Math Equations](#)

- | | |
|-----------------------|-----|
| ☐ | 7 |
| ☐ | 125 |
| ☐ | 127 |
| ☐ | 130 |

correct

MC180404545: SABEEHA AKRAM

MTH632:Grand Quiz

Question # 30 of 30 (Start time: 08:50:56 AM, 01 July 2020)

Let $z = 1 + i$, then $\arg(z) =$ _____.

Select the correct option

☐	$2\pi/4$
☐	$3\pi/4$
☐	$4\pi/4$
☐	$\pi/4$

correct



MC180404545: SABEEHA AKRAM

MTH632: Grand Quiz

Question # 29 of 30 (Start time: 08:50:06 AM, 01 July 2020)

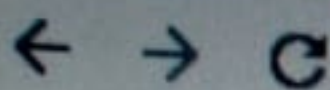
If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

Select the correct option

☒	analytic function
☐	need not be analytic function.
☐	analytic function only at $Z=0$.
☐	None of these.

correct





MC180404545: SABEEHA AKRAM

MTH632:Grand Quiz

Question # 6 of 30 (Start time: 08:18:36 AM, 01 July 2020)

If $-8i = 8 \exp[i(-n/2 + 2kn)]$, then the root offor $k=0$ is _____

Select the correct option

☒	$\sqrt{3} - i$
☐	$\sqrt{3} + i$
☐	$2(\sqrt{3} - i)$
☐	$2(\sqrt{3} + i)$

correct



MC180404545: SABEEHA AKRAM

MTH632: Grand Quiz

Question # 5 of 30 (Start time: 08:17:11 AM, 01 July 2020)

Harmonic conjugate of $u(x, y) = e^y \cos x$ is _____

Select the correct option

☒	$e^y \sin x + c$
☐	$-e^y \sin x + c$
☐	$e^x \sin y + c$
☐	$e^x \cos y + c$

correct

The Jordan Curve Theorem guarantees that a simple closed curve must enclose a region.

Select the correct option

- | | |
|-----------------------|----------------|
| ☐ | Euler curve |
| ☐ | Jordan curve |
| ☐ | Simple curve |
| ☐ | Rotation curve |

correct

Click to Save Answer & Move to Next Question

MC180404580: NAVEED AHMED

Time Left 85 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 18 of 30 (Start time: 08:30:20 AM, 01 July 2020) Total Marks: 1

Mapping is _____ way of visualizing a given function.

Select the correct option

- ☐ an analytical
- ☒ a geometrical
- ☐ a numerical
- ☐ none of the above

correct

Click to Save Answer & Move to Next Question

Question # 12 of 30 (Start time: 08:26:22 AM, 01 July 2020)

Total Marks:

A complex power function is a function of the form _____, where

 α

is a complex constant.

Select the correct option

[Reload Math Equations](#)☐

$$f(z) = z^{\alpha}$$

correct

☐

$$f(z) = \alpha^z$$

☐

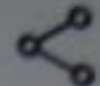
$$f(\alpha) = z^{\alpha}$$

☐

$$f(\alpha) = \alpha^z$$



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MC180404292: SHUMAILA KHAIR MUHAMMAD

Time Left 84 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 07:37 AM, 01 July 2020

Question # 30 of 30 (Start time: 08:07:52 AM, 01 July 2020)

Total Marks: 1

A point at which a function ceases to be analytic is called a _____ point.

Select the correct option



regular



non-regular



singular



non-singular

correct

Click on Each Answer & View of Next Question

MC180404580: NAVEED AHMED

Time Left88 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 1 of 30 (Start time: 08:10:46 AM, 01 July 2020)

Total Marks: 1

Let $z = 1 + i$, then $r =$ _____.

Select the correct option

 Reload Math Equations

☐	$\sqrt{2}$	correct
☐	$\sqrt{3}$	
☐	2	
☐	1	

Click to Save Answer & Move to Next Question



MC180404580: NAVEED AHMED

Time Left85 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 13 of 30 (Start time: 08:24:31 AM, 01 July 2020)

Total Marks: 1

If $f(z) = z^2$, $g(z) = \frac{1}{z}$, then $g(f(4)) =$

Select the correct option

Reload Math Equations

☐	4
☐	16
☐	1/4
☐	1/16

correct

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MC180404580: NAVEED AHMED

Time Left 86 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 7 of 30 (Start time: 08:18:18 AM, 01 July 2020) Total Marks: 1

If $f(z)=z^2$, then $f(x+iy)=$ _____.

Select the correct option

Reload Math Equations

☐	$x^2 - y^2 + 2xyi$	correct
☐	$x^2 + y^2 - 2xyi$	
☐	$x^2 + y^2i - 2xyi$	
☐	$x^2 - y^2i - 2xyi$	

Click to Save Answer & Move to Next Question



MC180404580: NAVEED AHMED

Time Left 87 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 11 of 30 (Start time: 08:22:07 AM, 01 July 2020) Total Marks: 1

If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____ in D.

Select the correct option

- | | |
|-----------------------|-------------|
| ☐ | equal |
| ☐ | steady |
| ☐ | harmonic |
| ☐ | homogeneous |
- correct

Click to Save Answer & Move to Next Question

Let $z = 7i$, then $r =$ _____.

Select the correct option

- | | |
|-----------------------|---|
| ☐ | 1 |
| ☐ | 2 |
| ☐ | 4 |
| ☐ | 7 |
- correct

Click to Save Answer & Move to Next Question

MC180404580: NAVEED AHMED

Time Left81 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 9 of 30 (Start time: 08:20:00 AM, 01 July 2020)

Total Marks: 1

If Cauchy – Riemann equations are not satisfied at a point z_0 then function is not _____ at z_0 .

Select the correct option

Reload Math Equations

☐	integrable
☐	differentiable

correct

Click to Save Answer & Move to Next Question

MC180404580: NAVEED AHMED

Time Left86 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 5 of 30 (Start time: 08:15:39 AM, 01 July 2020)

Total Marks: 1

Consider $\lim_{z \rightarrow z_0} f(z) = A$ and $\lim_{z \rightarrow z_0} g(z) = B$. Then $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} =$ _____.

Select the correct option

Reload Math Equations

☐	$\frac{A}{B}, B \neq 0$	correct
☐	$A + B$	
☐	$A - B$	
☐	$A \cdot B$	

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MC180404580: NAVEED AHMED

Time Left85 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 3 of 30 (Start time: 08:13:17 AM, 01 July 2020)

Total Marks: 1

The Cauchy - Riemann equations on a pair of real - valued functions of two real variables $u(x, y)$ and $v(x, y)$ are $U_x = V_y$ and _____.

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--------------|
| ☐ | $U_y = -V_y$ |
| ☐ | $U_x = -V_x$ |
| ☒ | $U_y = -V_x$ |
| ☐ | $U_x = -V_y$ |

correct

Click to Save Answer & Move to Next Question



Sum of complex numbers $(3 + 5i)$ and $(4 - 3i)$ is _____.

Select the correct option

- ☐ $1+5i$
- ☐ $7-3i$
- ☐ $7+5i$
- ☐ $7 + 2i$

correct

Click to Save Answer & Move to Next Question

MC180404580: NAVEED AHMED

Time Left 89 sec(s)

MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 2 of 30 (Start time: 08:12:08 AM, 01 July 2020) Total Marks: 1

If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(i)) =$

Select the correct option

Reload Math Equations

☒	i	correct
☐	i^2	
☐	i^3	
☐	-1	

Click to Save Answer & Move to Next Question